

Executive Report: SurplusSense Decision Support

SMU MBA Machine Learning — Individual Project | April 2026

1. Executive Summary

SurplusSense helps food merchants convert end-of-day surplus uncertainty into faster, safer, and more margin-aware decisions. It is a merchant-side decision-support cockpit — not a consumer marketplace — that translates surplus predictions into specific recommended actions (hold, monitor, discount, deep discount, donate, discard) under food-safety guardrails. Singapore generated 784,000 tonnes of food waste in 2024 (NEA); the merchant-side decision at 4pm is where that value is lost or recovered. No existing platform helps merchants decide what to discount, at what intensity, or when. SurplusSense fills this gap. The product is pilot-ready; commercial deployment requires a structured 4–6 week merchant pilot with 2–5 outlets.

2. Problem Worth Solving

For F&B merchants — bakeries, cafés, and prepared-food outlets — the problem is a daily operational decision at 4pm: **what do I do with unsold inventory?**

The decision is genuinely hard:

- **Time pressure:** shelf life is already running down
- **Margin trade-off:** deeper discounts recover more revenue but erode margin on items that would have sold at full price
- **Safety risk:** items past safe holding times cannot be listed
- **Labour constraints:** staff must decide under operating pressure
- **Inconsistent judgment:** the same surplus item may receive different decisions across shifts

The monetisable gap is narrower than "food waste" broadly: merchants often know they may have surplus, but lack a structured way to decide what action to take before the item loses value. This is not only a forecasting problem — it is a time-sensitive trade-off between revenue recovery, food safety, staff workload, and customer experience.

Target buyer: Singapore-based bakery, café, or prepared-food outlet with recurring end-of-day surplus and enough daily volume for recovered value to exceed SGD 99–299 monthly subscription.

3. Product Approach

SurplusSense is a **decision-support product** — it translates surplus-risk predictions into specific merchant actions. The six-step workflow:

Step	What happens
1 — Enter context	Merchant selects product category, storage type, holding time, shelf life

Step	What happens
2 — Predict surplus	XGBoost model estimates expected surplus units from 47 engineered features
3 — Recommend action	Rule engine maps surplus quantity and shelf life to HOLD / MONITOR / DISCOUNT / DEEP DISCOUNT / DONATE / DISCARD
4 — Screen for safety	Five deterministic checks return SAFE / CAUTION / BLOCKED
5 — Estimate recovery	System calculates discounted revenue recovery vs potential loss
6 — Explain recommendation	Merchant sees the specific reason for the recommended action

Why rules over learning: Food safety cannot be entrusted to an opaque learned policy. A merchant who lists an unsafe item creates liability that no accuracy metric offsets. Rule-based logic is fully explainable — every recommendation traces to a specific rule. The model estimates what *may* happen; the rule and safety layers decide what action is *permissible and commercially sensible*.

4. ML Architecture and Results

Architecture:

Layer	Component	Role
Input	Merchant/item context	31 raw features: temporal, lag, rolling, expanding-window, product
Feature engineering	Expanding-window aggregates with shift(1)	Prevents target leakage; no future information in aggregates
Prediction	XGBoost	Estimates surplus units
Validation	Temporal 80/20 holdout + TimeSeriesSplit CV	Forward-looking evaluation; simulates real deployment
Decision	Rule-based recommendation engine	Converts surplus prediction into action
Governance	Food-safety rules	BLOCK/CAUTION/SAFE overrides commercial optimisation
Output	Merchant decision cockpit	Recommended action + recovery estimate + explanation

Key design decisions:

Decision	Choice	Why
Train/test split	Temporal holdout	Random split would inflate accuracy using future information

Decision	Choice	Why
Feature aggregation	Expanding-window with shift(1)	Prevents target leakage; each row only uses data before its date
Recommendation logic	Transparent rule engine	Food safety requires explainable logic; black-box learned policy rejected
New merchant handling	Category benchmark fallback	Makes product usable from day one without historical data
Output format	Recommended action + recovery value	Matches real merchant decision workflow, not a prediction score

Model results (synthetic data, 4,027 records, 15 merchants, 13 categories, 90-day window):

Model	Temporal Holdout MAE	vs Historical Average
Historical Average	1.49	—
Previous Day	2.01	−35%
XGBoost (Tuned)	0.6355	−57%

5-fold TimeSeriesSplit CV MAE: 1.38 ± 1.22 (higher variance reflects distributional shift across temporal folds).

5. Business Case and Pilot Validation

Who pays: Bakeries, cafés, and small F&B operators with 5–50 daily surplus units in Singapore.

Why they pay: Recovered surplus value plus avoided disposal costs could exceed SGD 99–299 per month per outlet, subject to pilot validation. Even SGD 6–12 daily recovery can justify the entry subscription tier if staff use the system consistently.

Merchant unit economics:

Scenario	Daily recovered value	Monthly (26 operating days)	Supports pricing
Conservative	SGD 6/day	SGD 156/month	Entry plan SGD 99/month
Base case	SGD 12/day	SGD 312/month	Mid plan SGD 199/month
Strong case	SGD 20/day	SGD 520/month	SGD 299/month + transaction fee

Proposed pricing: Monthly SaaS SGD 99–299 per outlet (predictable recurring revenue); 15% recovered-value fee (aligns SurplusSense revenue with merchant benefit); ESG/reporting module for chains.

Pilot design (required before commercial deployment claims):

- **Duration:** 4–6 weeks with 2–5 merchant outlets
 - **Data:** Daily sales, inventory, discount, donation, and disposal records from merchant POS/inventory systems
 - **Success metrics:** Recommendation acceptance rate $\geq 60\%$; surplus reduction 10–15% vs baseline; recovered value $\geq$ SGD 200/outlet/month; zero safety incidents
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SurplusSense is a pilot-ready decision-support prototype. It is not SFA-validated and not commercially live. All performance metrics are from synthetic data and require validation with merchant POS/inventory data before commercial deployment claims.

Source note: Singapore generated 784,000 tonnes of food waste in 2024 (NEA waste statistics). Business impact estimates are derived from synthetic pilot data and model outputs; real merchant pilot data is required before commercial rollout.